Stage 10. Baseline

The final backup and its restore verification

PREPARED BY Energy Hippo implementation team
STATUS Ready to enter

Stage 10 closes the implementation out. Everything from Stage 1's companies through Stage 9's report delivery has been entered through the applications, every stage's checks are green, and now the whole thing is captured in one backup that can be trusted: restored anywhere, it reproduces Northlake exactly, not "approximately." This stage takes that backup, proves it by restoring it somewhere disposable and reading real values back out, and then promotes it to be the reference database everyone restores from afterward.

Before you start

- Stages 1-9 are green: every checks.sql in every stage folder has been run and matches its expected result.
- This stage is SQL Server work, not an EEM editor. Run its commands with sqlcmd or SQL Server Management Studio as a sysadmin, the same way Stage 3's entry backup was taken.
- A database backup only ever captures the database. It does not capture the application version, the repo state, or anything held outside SQL Server (gateway folders, ConfigFiles runtime settings, external service credentials). Step 1 records all of that beside the .bak file.

Step 1. Final backup

Where: SQL Server, `BACKUP DATABASE`.

Enter: 01-backup.yaml.

Order: confirm every stage's checks are green, then take one full backup of `EEMSuite_Northlake` to `C:\Hippo\Backups\`, named `EEMSuite_Northlake.stage10-final.<yyyyMMdd>.bak` — the same folder and dot- separated naming the existing pre-upgrade backup already uses there, distinct from Stage 3's entry backup in the instance's own default backup path. Run `RESTORE VERIFYONLY` against it immediately.

Keep beside it, since none of it is in the database: the application version (read it from the installed app; the tree builds 7.1, and 7.0.0.4005 in earlier drafts is stale) and the repo commit the install was built from; the gateway input/working/backup/error folders and `GatewayOutput` state files under `C:\ProgramData\EEM Suite\NorthLake`; and any externally-held settings — ConfigFiles runtime configuration, the ENERGY STAR `CustomerServer` constant and Portfolio Manager credentials once connected, SMTP/mail relay settings.

Record: the backup file path and the date taken.

Check: `backup_history`.

Step 2. Restore verification

Where: SQL Server, `RESTORE DATABASE`, then the web site and Database Administrator pointed at the restored copy.

Enter: 02-restore-verification.yaml.

Order: restore the Step 1 backup into a disposable database (`EEMSuite_NorthlakeVerify`) on the same server, never over the real `EEMSuite_Northlake`. Point gateway profiles at scratch folders instead of the real ProgramData paths, and leave the ENERGY STAR and mail settings unset, since this is a read/render check, not a live-processing rehearsal. Then open the report checklist in the file — one representative report per stage — and confirm each shows the value it should.

Once every row in the checklist matches: the Step 1 backup is promoted. It replaces the 2026-09-14 pre-upgrade snapshot as Northlake's reference database, the file every future restore starts from. Drop `EEMSuite_NorthlakeVerify` and its scratch folders; the real instance was never touched.

Check: `baseline_counts`, `ridgeline_bill_values` and `activity_and_audit_survive_restore`. Each returns an expected value, the actual value and PASS or FAIL, and the same values must come back from the restored copy as from the original. `checklist_not_scripted` lists the checklist lines that can only be read by eye.

Stage 10 is done when

- The Step 1 backup exists in `msdb` backup history as a full backup at `C:\Hippo\Backups\`, and `RESTORE VERIFYONLY` passed.
- Every row of `baseline_counts` is PASS: the three named companies, six providers, 13 ENERGY STAR properties, 4 regressions, 3 GHG configurations, 5 projects, 15 alarm points, 23 saved-report rows, 3 dashboards, 5 deliveries with 20 recipient rows and no EnergyAI loads.
- `ridgeline_bill_values` is PASS: the July 2024 statement at 129,330.27, the generator's first bill at 1,077.45 and the -152,792.65 credit on the one Central Plant bill and no other.
- The full report coverage matrix's 119 reports have each rendered at least once; the matrix records the value a stage expects for each, where one is known.
- The restore-verification checklist opened cleanly in a disposable database and every value matched.
- The backup is promoted as Northlake's reference database.

Run `checks.sql` beside this chapter against the instance database to see each of these as a result set.

Findings

None new in this stage. It inherits every finding recorded in Stages 8 and 9 (ENERGY STAR scores are external, benchmark and MFR external-email UI gaps, EnergyAI and GreenButton have no entry path, custom report registration has no UI) — see product findings for the full list. Nothing in Stage 10 works around any of them with SQL; the restore verification checklist expects those reports to show exactly the partial state those findings describe (property metrics without a score, a delivery reaching people only through groups, and so on), not a hidden seed of the missing data.